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Comp 3004 Assignment 3

Use case

Name : Normal use of elevator

Primary Actor: Passenger, Elevator, ECS (Elevator Control System)

Stakeholders and Interests:

Building safety – Deals with elevator malfunction situations.

Building – Sends fire alarm signal

Pre-Conditions: Passenger wants to take the elevator.

Success guarantee: Passenger safely arrives on the desired floor.

Main success scenario:

1. Passenger presses a floor button to call the elevator.
2. Floor button illuminates.
3. Floor button informs ECS about the request from the passenger.
4. ECS sends floor request to elevator.
5. Elevator updates destination.
6. Elevator moves towards destination.
7. Elevator arrives and turns off button light.
8. Elevator rings a bell.
9. Elevator opens its doors.
10. Passenger enters elevator.
11. Elevator door closes after 10 seconds.
12. Passenger select one or more destination floors using a panel of buttons.
13. Elevator updates destination.
14. Elevator moves towards destination.
15. Elevator arrives at a destination floor and rings the bell.
16. Elevator opens door.
17. Elevator door closes after 10 seconds.

Steps 13 – 17 are repeated until elevator is going to the last selected floor or passenger is arriving at destination floor.

18. Elevator arrives at a destination floor and rings the bell.
19. Elevator opens door.
20. Passenger walks out of the elevator.
21. Elevator door closes after 10 seconds.
22. The use case is complete.

Extensions:

*a. Elevator detects a floor sensor:

*a1. Elevator displays updated current floor on display and continue.

*b. "Help" button is pressed:

*b1. Use case "Help alarm" is called.

*c. Light sensor on the door is interrupted when the door is closing.

*c1. Use case "Door obstacles" is called.

*d. Elevator receives fire signal from building:

*d1. Use case "Fire" is called.

*e. Elevator detects fire inside elevator:

*e1. Elevator sends fire signal to building.

*e2. Use case "Fire" is called.

*f. Sensors indicate that the load exceeds the carrying capacity:

*f1. Use case "Overload" is called.

*g. Elevator receives a "Power Out" alarm signal:

*g1. Use case "Power Out" is called.

11a. Passenger presses the "open door" button:

11a1. Elevator doors remain open until passenger releases the button.

11a2. Repeat step 11.

11b. Passenger presses the "close door" button:

11b1. Elevator doors close immediately and continue.

17a. Passenger presses the "open door" button:

17a1. Elevator doors remain open until passenger releases the button.

- 17a2. Repeat step 18.
- 17b. Passenger presses the “close door” button:
 - 17b1. Elevator doors close immediately and continue.
- 21a. Passenger presses the “open door” button:
 - 21a1. Elevator doors remain open until passenger releases the button.
 - 21a2. Repeat step 22.
- 21b. Passenger presses the “close door” button:
 - 21b1. Elevator doors close immediately and continue.
- 22a. Elevator has not been to every selected floor:
 - 22a. Repeat step 13 – 17 until it has been to every selected floor.

Name: Help alarm

Primary Actor: Passenger, Elevator, building safety.

Stakeholders and Interests: Emergency services – to receive emergency calls.

Pre-Conditions: Passenger pressed the “Help” button.

Success guarantee: Passenger manages to speak to building safety.

Main success scenario:

1. Elevator connects to building safety with the mic.
2. Building safety responds to the call.
3. Passenger responds to the call.

Extensions:

- 2a. Building safety fails to respond to the call in 5 seconds.
 - 2a1. 911 emergency call is placed.
 - 2a2. Use case is terminated.
- 3a. Passenger fails to respond to the call.
 - 3a1. 911 emergency call is placed.
 - 3a2. Use case is terminated.

Name: Door obstacles

Primary Actor: Elevator, Passenger

Stakeholders and Interests: none

Pre-Conditions: Light sensor is interrupted when the door is closing.

Success guarantee: Passenger removes item obstructing the door.

Main success scenario:

1. Elevator stops the door from closing.
2. Elevator opens the door.
3. Passenger removes item obstructing the door.
4. Elevator closes the door and continues.

Extensions:

*a. Door is obstructed repeatedly in a short period of time:

*a1. Elevator plays a warning audio over the warning system.

*a2. Elevator displays a text message and continues.

Name: Fire

Primary Actor: Elevator, Passengers

Stakeholders and Interests: ECS – provide floor information.

Building safety – receives information from elevator.

Pre-Conditions: Elevator received a fire alarm signal (from building or itself)

Success guarantee: Elevator successfully reached a safe floor and released the passengers.

Main success scenario:

1. Elevator plays a warning audio telling passengers to disembark once on a safe floor.
2. Elevator plays displays warning texts telling passengers to disembark once on a safe floor.
3. Elevator goes to a safe floor.
4. Elevator opens the doors.

5. Passengers leave the elevator.

Extensions:

3a. Elevator is unable to get to a safe floor:

3a1. Elevator informs building safety.

3b2. Elevator informs ECS to find a new safe floor.

3b3. Repeat step 3.

Name: Overload

Primary Actor: Elevator, Passengers

Stakeholders and Interests: none

Pre-Conditions: Sensors indicated that the load exceeds the carrying capacity.

Success guarantee: The load is reduced to lower than the carrying capacity.

Main success scenario:

1. Elevator stops moving.
2. Elevator opens door.
3. Elevator plays a warning audio over the warning system.
4. Elevator displays a text message.
5. Passengers step out of the elevator.

Steps 3 – 5 are repeated until the load is under the carrying capacity.

6. Continues.

Extensions: none

Name: Power Out

Primary Actor: Elevator, Passengers

Stakeholders and Interests: ECS – provide floor information.

Building safety – receives information from elevator.

Pre-Conditions: Elevator received a “Power Out” alarm signal.

Success guarantee: Elevator arrived at a safe floor and released the passengers.

Main success scenario:

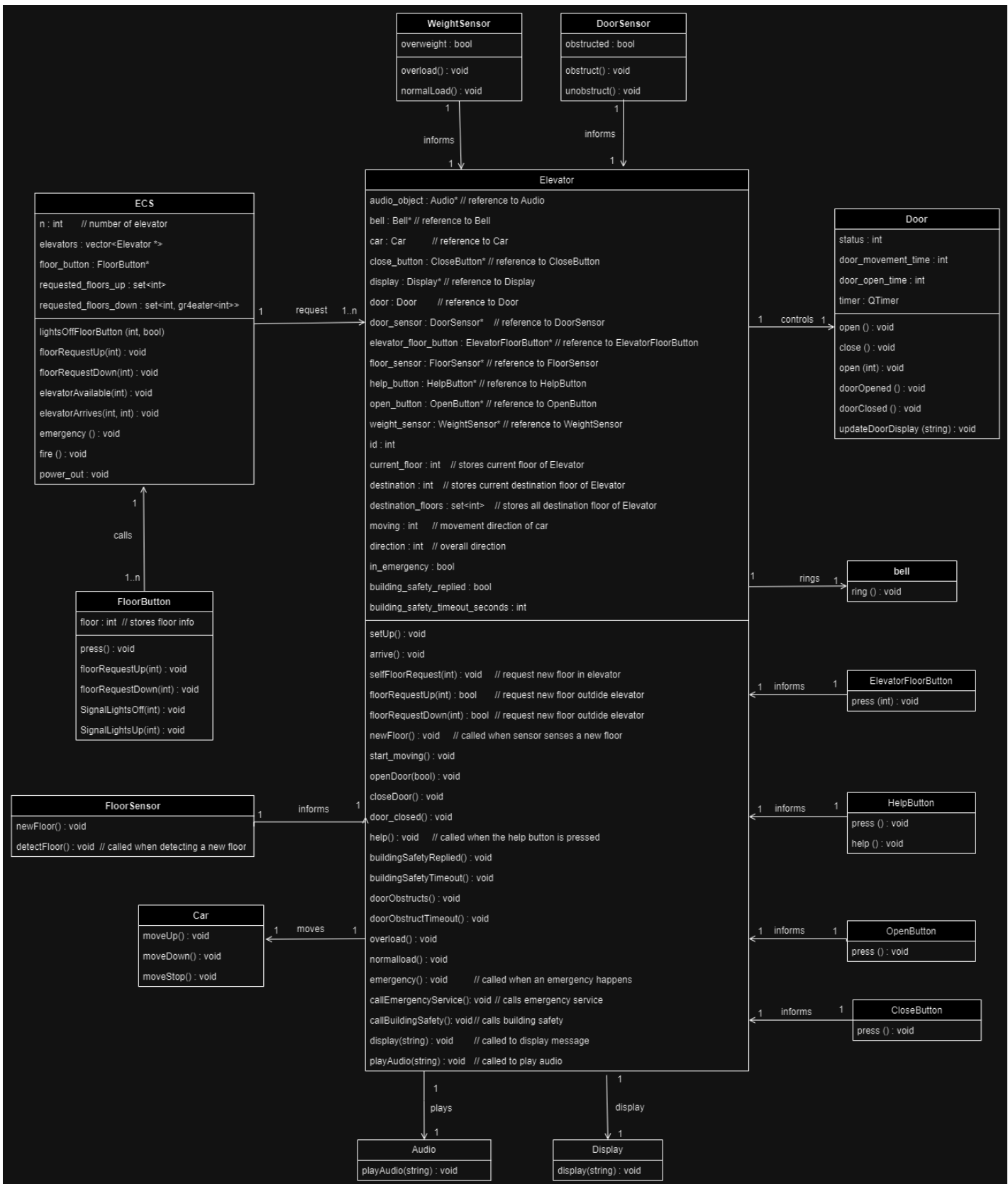
1. Elevator plays a warning audio telling passengers to disembark once on a safe floor.
2. Elevator plays displays warning texts telling passengers to disembark once on a safe floor.
3. Elevator goes to a safe floor.
4. Elevator opens the doors.
5. Passengers leave the elevator.

Extensions:

3a. Elevator is unable to get to a safe floor:

- 3a1. Elevator informs building safety.
- 3b2. Elevator informs ECS to find a new safe floor.
- 3b3. Repeat step 3.

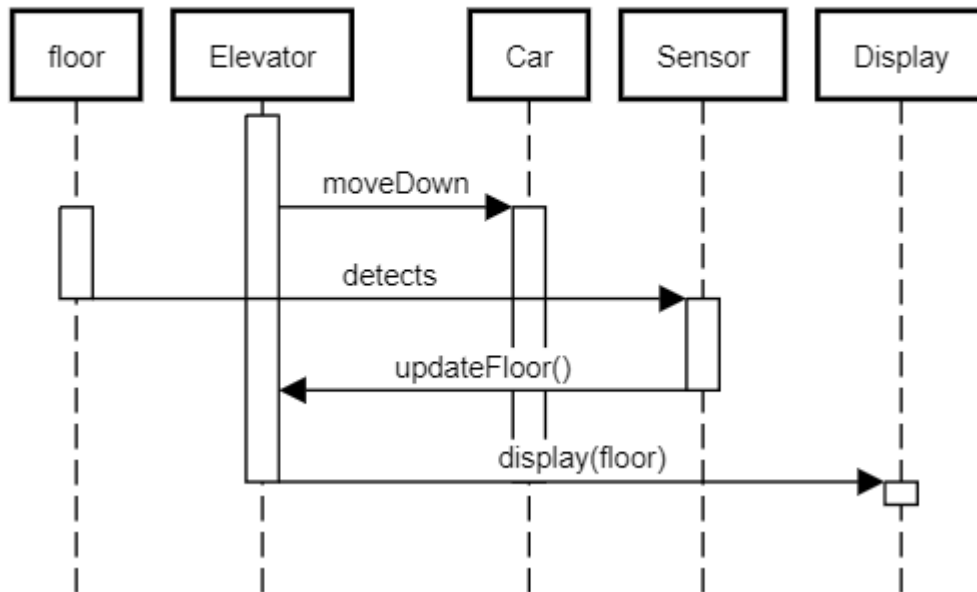
UML Class diagram



sequence Diagram

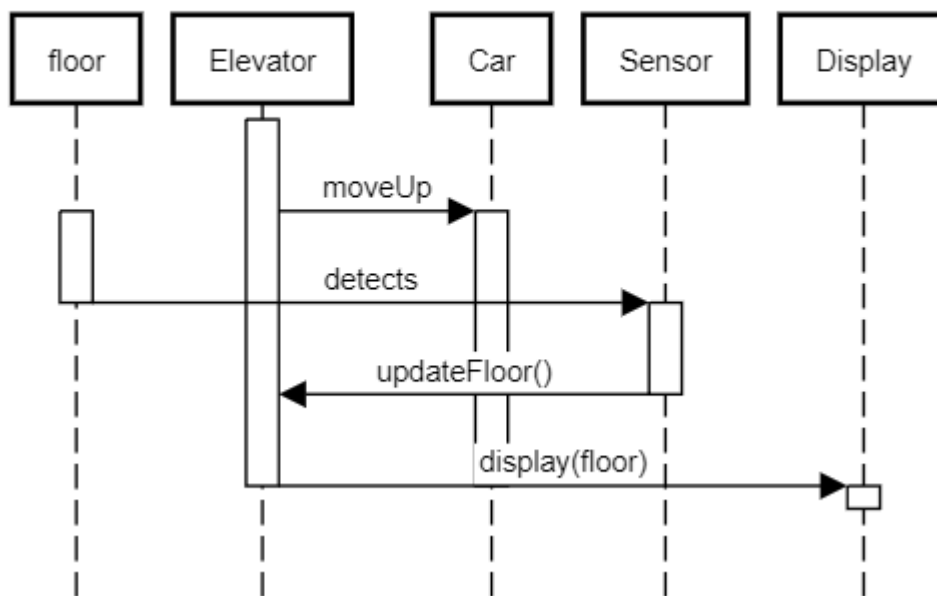
elevator moving up

Elevator Moving Down

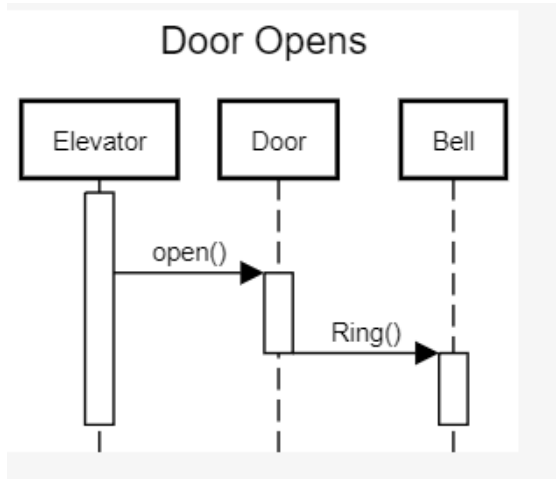


elevator moving down

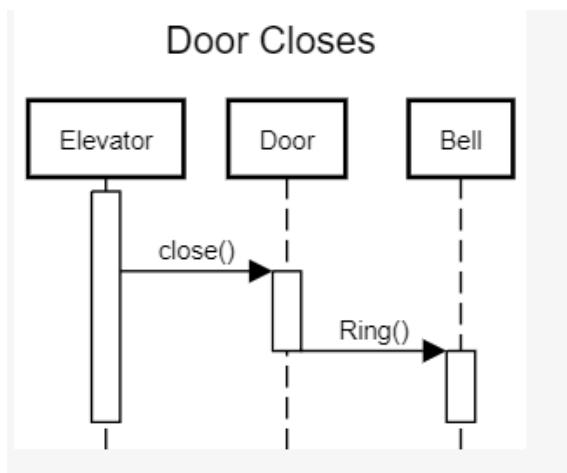
Elevator Moving Up



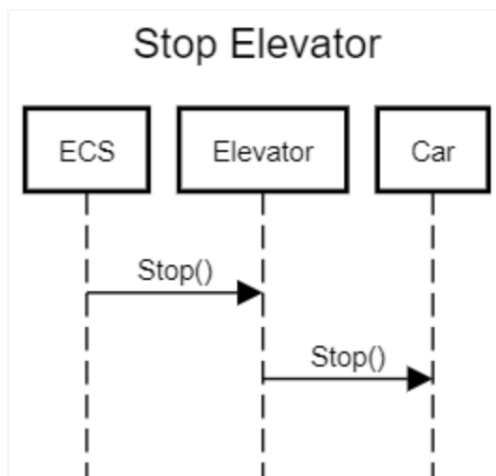
Door Opens



Door Closes



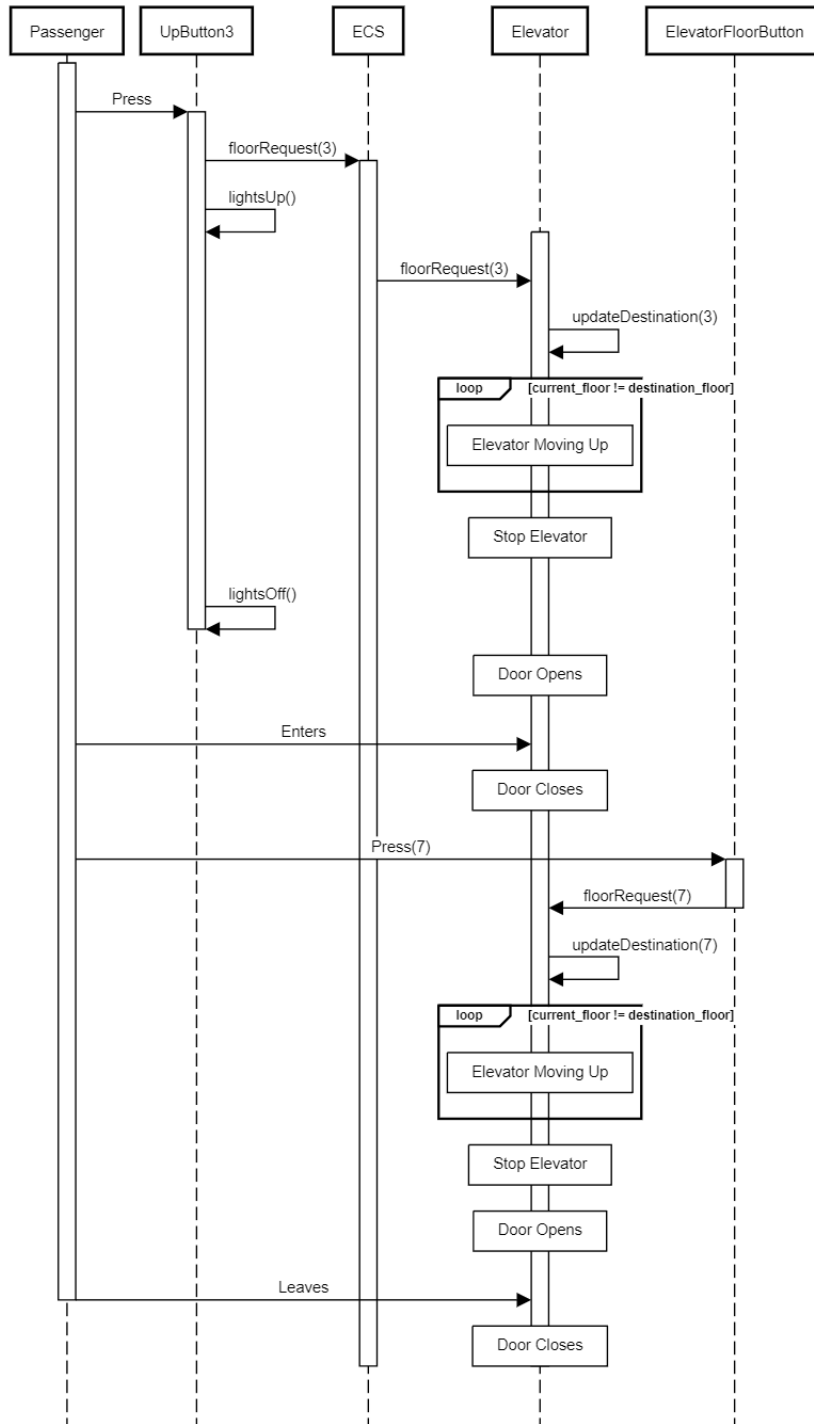
Stop Elevator:



success scenario 1: 1 passenger going from floor 3 to floor 7

1 elevator in total, positioned at floor 1

Passenger goes from floor 3 to floor 7



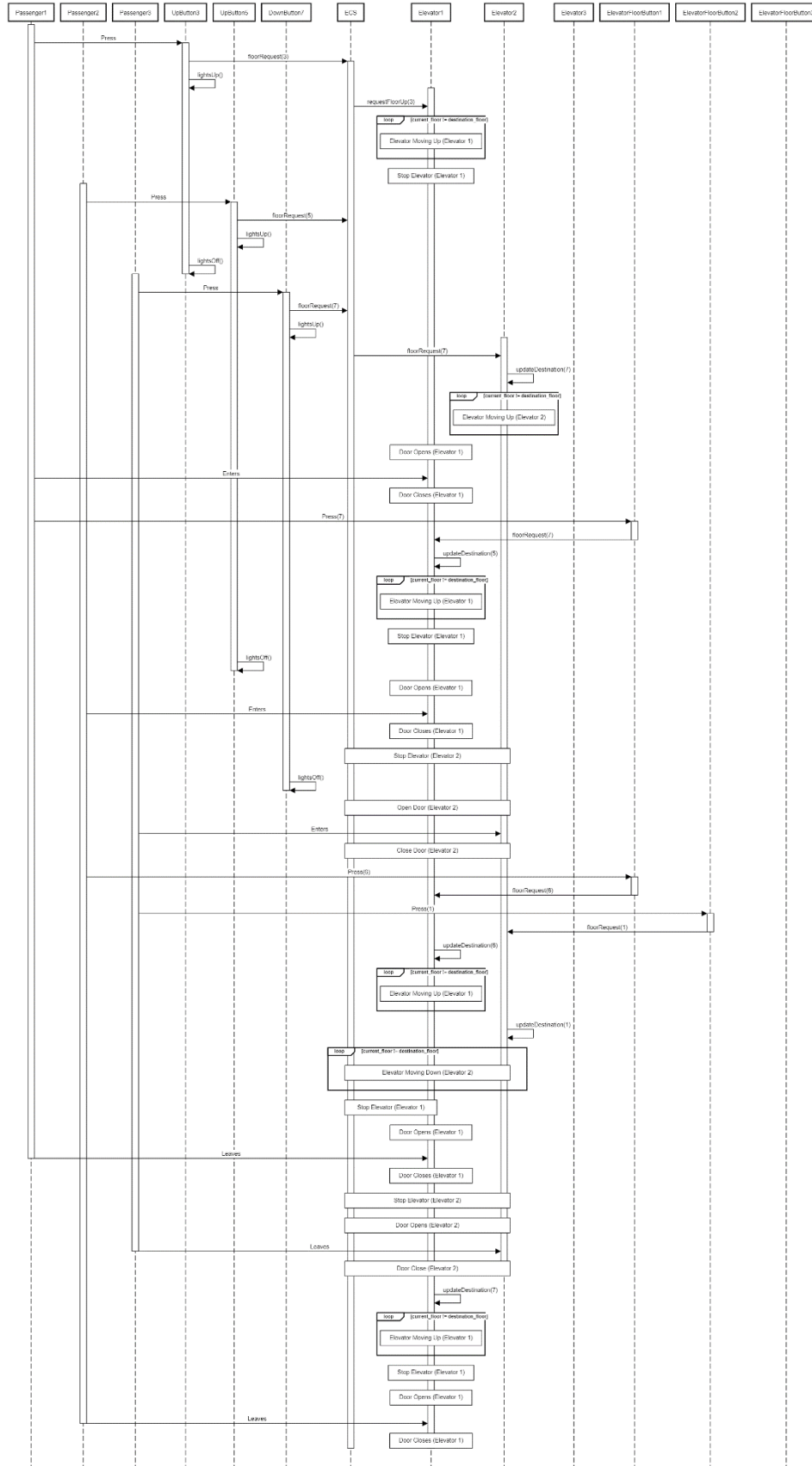
Success scenario 2: 3 passengers, 3 elevators

Passenger 1: goes from 3rd to 7th floor

Passenger 2: goes from 5th to 6th floor

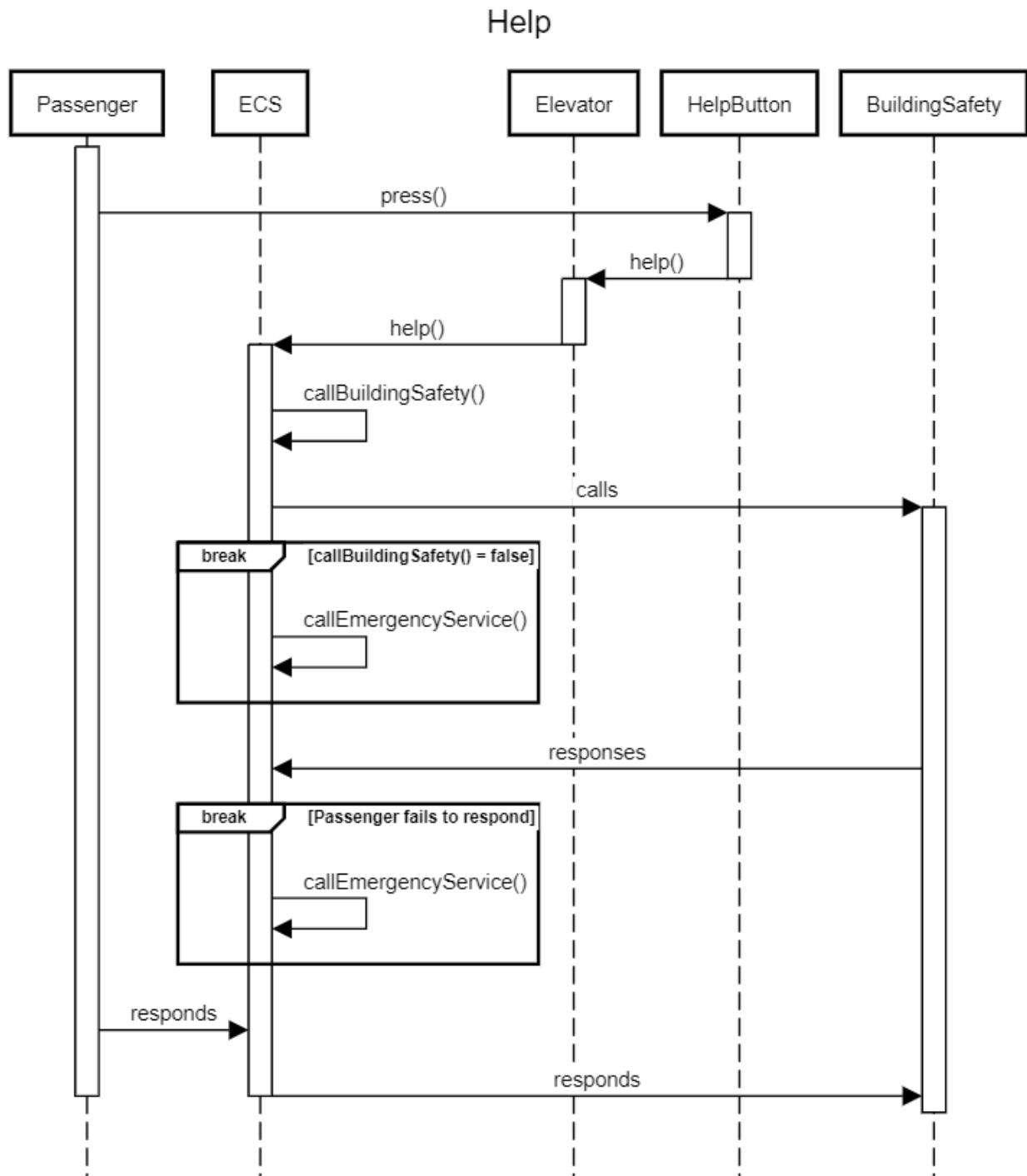
Passenger 3: goes from 7th to 1st floor

Passenger1 goes from floor 3 to floor 7

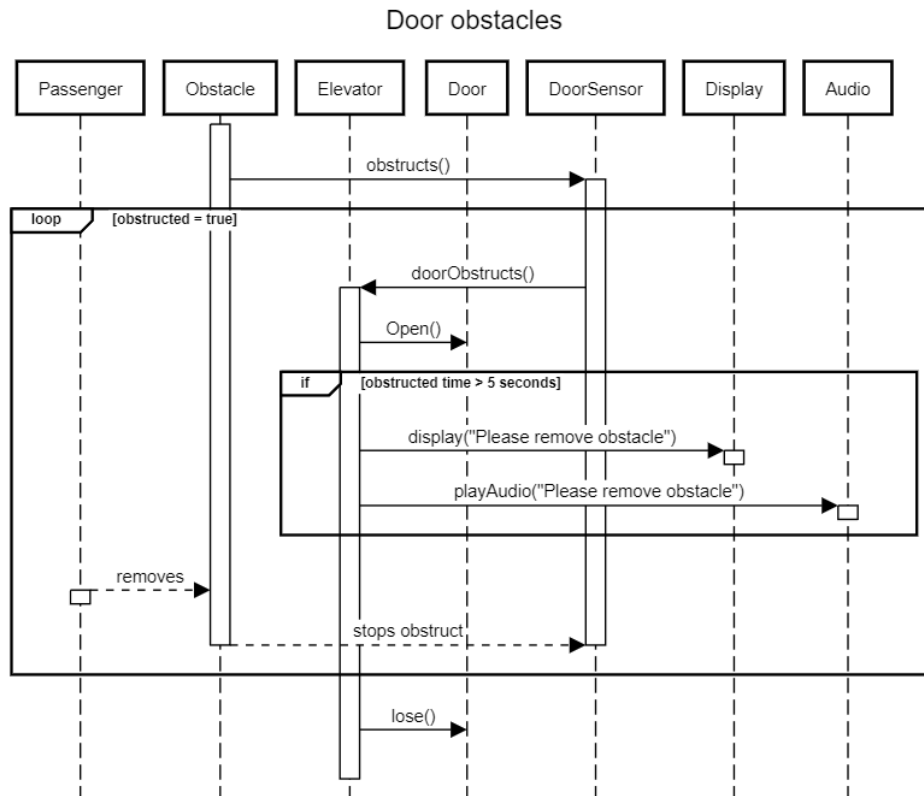


Safety Scenarios

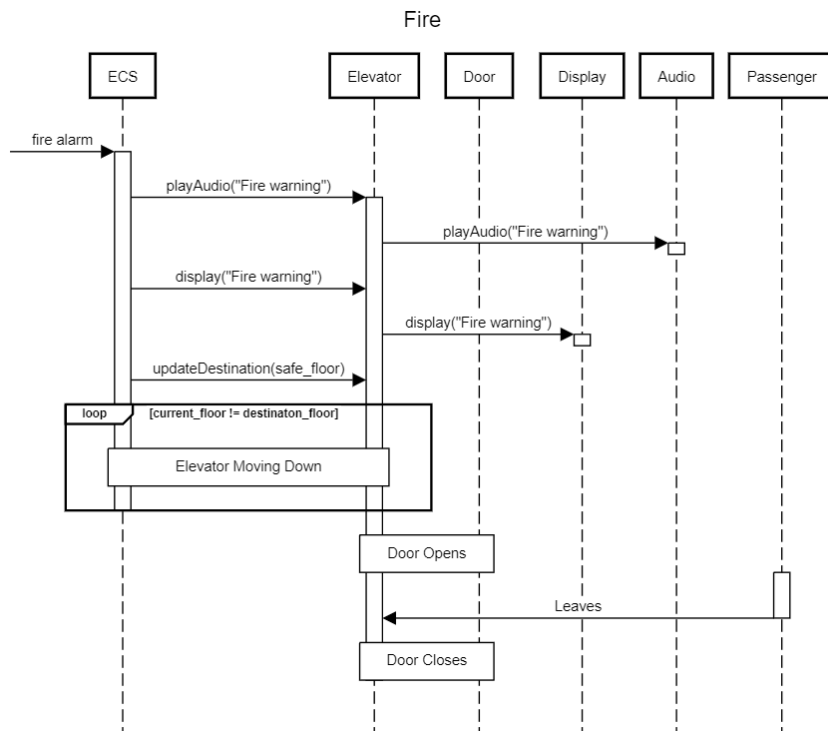
Help



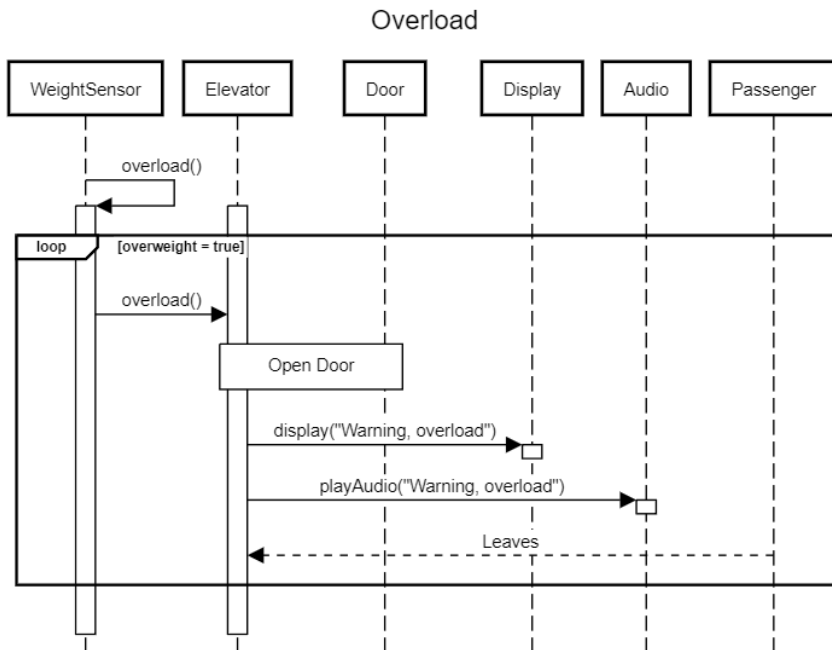
Door Obstructs



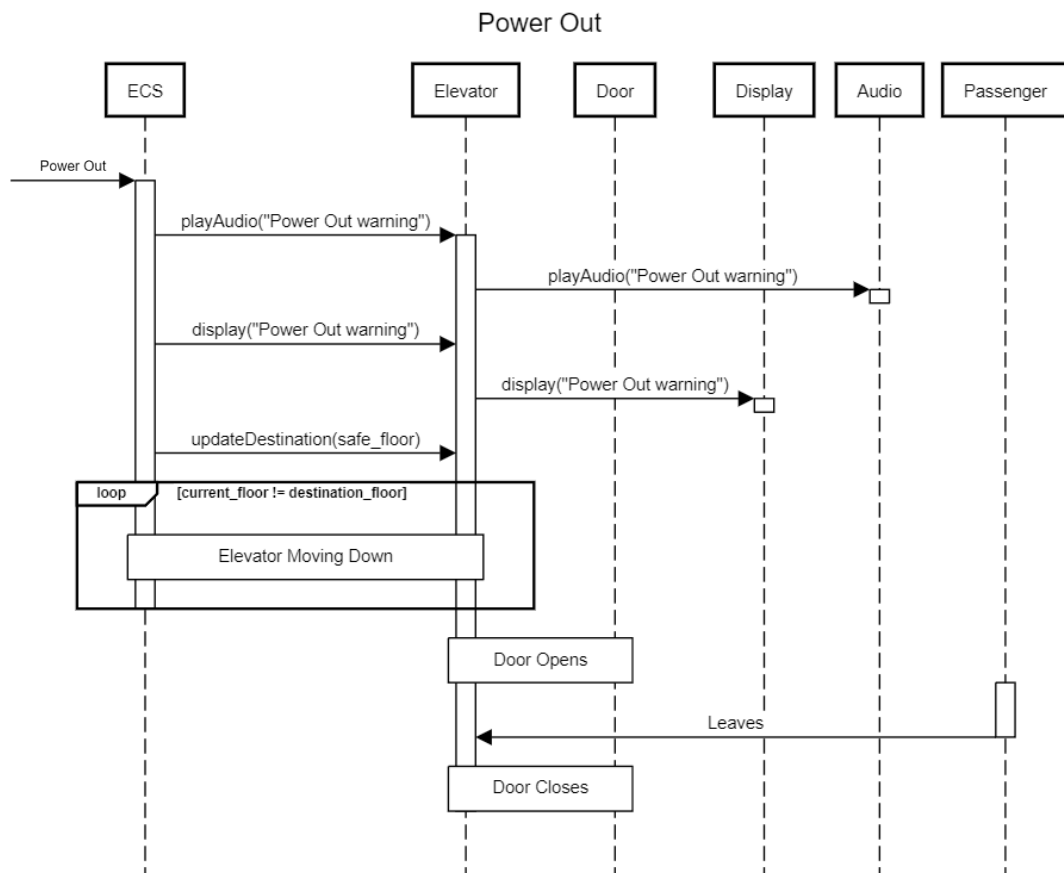
Fire



Overload

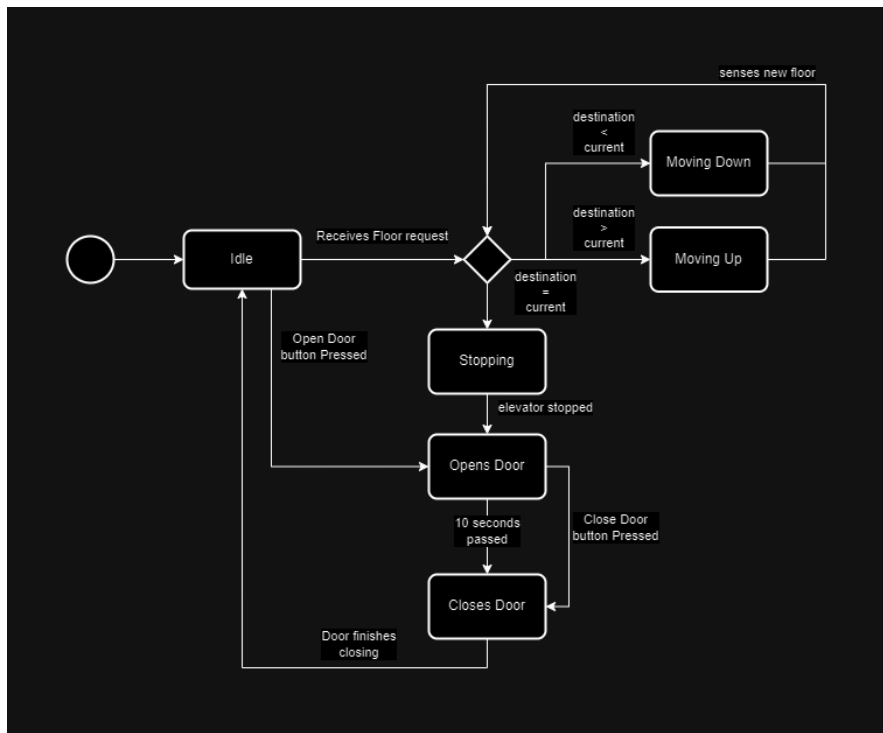


Power Out



State Diagram

decentralized



Discussion

I chose to follow the Decentralized design as all the elevators will decide where and how to move on their own. A centralized ECS is used to pass requests to all elevators and store requests when all elevators are busy. It also receives emergency signals and passes it to all elevators.

Requirements Traceability Matrix

id	Requirement	Fulfilled by	Tested by	Description
1	A building is serviced by M elevators (also called cars). On each of the N floors is a pair of buttons marked “up” and “down”.	mainwindow.h ecs.h elevator.h floor_button.h	Run the program and the window with all interface will display the elevators and floor buttons.	ECS class stores a list of elevator references and a floor button object reference. Main window has buttons and interfaces that represent the floor buttons and elevators.
2	When a button is pressed it illuminates, and remains illuminated	floor_button.h floor_button.h -press() -lightsUp() -lightsOff()	Press the floor buttons and the floor buttons will light up.	FloorButton has a press() function for when it is pressed and functions to turn on and off lights, and sends a floor request to ecs.
3	When the elevator arrives, it rings a bell, opens its doors (the elevator and floor doors) for a fixed time (10 seconds) allowing people to exit or board, rings the bell again, closes its doors	bell.h -ring() door.h -open() -close()	Move an elevator to a requested floor, and the prompt will show the bell is ringing and the door is opening or closing.	Elevator calls open and close functions in the door class. The door calls ring function in bell class when it opens and closes. Door sets a timer of 10 seconds and closes itself when timer is over.
4	Once on-board passengers select one or more destination floors using a panel of buttons; there is one button for every floor	elevator_floor_button.h -press(int) elevator.h -floor_request(int)	Press a floor button in the elevator interface, and the elevator will starting going to the destination floor.	When pressed, press function in Elevator Floor Button class is called with the floor pressed. Elevator Floor Button will call floor request function in Elevator class and Elevator will update it's destination and starts moving.

5	The elevator has a display which shows passengers the current floor of the elevator	elevator.h	Floor display shows the current floor of the elevator.	when elevator reaches a new floor, it calls the main window to update the display
6	There is also a pair of buttons on the elevator control panel marked "open door" and "close door". These buttons can be used by a passenger to override the default timing of the doors.	open_button.h close_button.h -press() door.h	Press the open and close button in elevator interface when elevator is not moving, the interface will show the status of the door.	Open button class and Close button class's press function is called when pressed, which will override the default timer and open / close the door.
7	Inside the elevator there is also a help button linked to building safety service.	help_button.h -press() elevator.h -help() -callBuildingSafety() - callEmergencyService()	There is a help button in each of the elevator interface.	When pressed, press function is called in help button class. Help function in the elevator class will be called by the help button class. Then the Elevator class will call its callBuildingSafety function and callEmergencyService function if needed.
8	Each elevator has a sensor that notifies it when it arrives at a floor. The elevator control system should ensure that the group of elevators services all (floor and on-board) requests expeditiously.	floor_sensor.h elevator.h -newFloor()	Move the elevator to a new floor, the floor display will update.	When the floor sensor senses a new floor, the detectFloor function is called. Then the floor sensor will call the newFloor function in the elevator class. Then the elevator will compute the new floor and update the floor.

9	Each elevator has a display and an audio system. The display shows the current floor number and warning messages that are synced with audio warnings.	display.h -display() audio.h -playAudio() elevator.h -display() -playAudio()	Perform actions where the display and audio will be updated (overload, emergency etc). The display and audio box will be updated.	When elevator class receives an emergency signal, the display and the playAudio function in the elevator class will be called to display and play a message.
10	The control system receives a "Help" alarm signal from an elevator indicating that the "Help" button has been pressed. In that case, the passenger is connected to building safety service through a voice connection. If there is no response from building safety within 5 seconds or if there is no response from a passenger a 911 emergency call is placed.	help_button.h -press() elevator.h -help() ecs.h -help() -callBuildingSafety() -callEmergencyService()	Press the help button in the elevator interface. The prompt will print that help is called.	When the help button is pressed the press function is called. Then the helpButton class will call the help function in elevator class. The Elevator will try to call the callBuildingSafety function, and if it returns false (no reply), it calls the callEmergencyService function.
11	If the light sensor is interrupted when the door is closing, the control system stops the door from closing and opens it. If this occurs	door_sensor.h -obstructs() elevator.h -doorObstructs() door.h -open()	Press the obstruct button in the elevator interface when the door is closing. Press the unobstruct button to remove the obstruct item.	When door sensor senses something, it calls the obstructs function. The function calls the doorObstructs function in the elevator.h, then the elevator class calls the open function in

	repeatedly over a short period of time, a warning is sounded over the audio system and a text message is displayed.			the door class to open the door.
12	The control system receives a "Fire" alarm signal from the building and commands all elevators to move to a safe floor. Similarly, a "Fire" alarm signal from the elevator itself will cause that elevator to go to a safe floor. In both cases an audio and text message are presented to passengers informing them of an emergency and asking them to disembark once the safe floor is reached.	ecs.h -emergency() elevator.h -emergency() -display() -playAudio() car.h -moveUp() -moveDown() -stop() door.h -open() display.h -display(string) audio.h -playAudio(string)	Press the fire button on the right hand side. The elevator will try to move to a safe floor and opens the door.	When a fire happens, the emergency function is called in ecs. The ecs process the request and calls the elevator to move to a safe floor. The elevator displays and plays a warning text. Upon arrival, the elevator opens the door for the passengers to leave.
13	The control system receives an "Overload" alarm signal from an elevator if the sensors indicate that the passenger or cargo load exceeds the carrying capacity. In that	weight_sensor.h -overload() elevator.h -overload() -display(string) -playAudio(string) door.h -open() display.h -display(string) audio.h -playAudio(string)	Press the overload button to simulate overloading, and the normal load button to remove the overload. When the door is not closed, the display box will show that the	When the weight sensor senses an overload, the overload function is called, which calls the overload function in the elevator class. The elevator class will call the open function in door class, and display

	case, the elevator does not move and an audio and a text messages are presented to passengers asking for the load to be reduced before attempting to move again.		elevator is overloaded, and the door is kept opened.	and play a warning text.
14	The control system receives a "Power Out" alarm signal. In that case, an audio and a text messages are presented to passengers informing them of the power outage. Each elevator is then moved to a safe floor and passengers are asked to disembark via audio and text messages.	ecs.h -emergency() elevator.h -moveDown() -stop() -display() -playAudio() car.h -moveUp() -moveDown() -stop() door.h -open() display.h -display(string)	Press the power out button on the right hand side. The elevator will try to move to a safe floor and opens the door.	When a power out happens, the emergency function is called in ecs. The ecs process the request and calls the elevator to move to a safe floor. The elevator displays and plays a warning text. Upon arrival, the elevator opens the door for the passengers to leave.